Emergency Airway Management During Awake Craniotomy: Comparison of 5 Techniques in a Cadaveric Model

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Background: During awake craniotomy, securing the patient's airway might be necessary electively or emergently. The objective of this study was to compare the feasibility of airway management using a laryngeal mask airway (LMA) and 4 alternative airway management techniques in an awake craniotomy simulation.

Methods: After completing a questionnaire, 9 anesthesia providers attempted airway management in a cadaver positioned to simulate awake craniotomy conditions. Following the simulation, participants rated and ranked the devices in their order of preference.

Results: Only 3 approaches resulted in the successful securement of an airway device for 100% of participants: LMA (median; interquartile range time to secure the airway 6 s, 5 to 10 s), fiberoptic bronchoscopy through an LMA (41 s; 23 to 51 s), and video laryngoscopy (49 s; 43 to 127 s). In contrast, the oral and nasal fiberoptic approaches demonstrated only 44.4% (154.5 s; 134.25 to 182 s) and 55.6% (75 s; 50 to 117 s) success rates, respectively. The LMA was the fastest and most reliable primary method to secure the airway (P=0.001). After the simulation, 100% of participants reported that an LMA would be their first choice for emergency airway management, followed by fiberoptic intubation through the LMA (7 of 9 participants) if the LMA failed to properly seat.

Conclusions: We demonstrated that an LMA was the fastest and most reliable primary method to secure an airway in a laterally positioned cadaver with 3-pin skull fixation. Fiberoptic and video laryngoscope airway equipment should be readily available during awake craniotomy procedures, and an attempt to visualize the vocal cords through the LMA should be attempted before removing it for alternative techniques.

Key Words: airway management, awake craniotomy, intubation, laryngeal mask airway, neurosurgical procedures, simulation training

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Awake craniotomy (AC) is a surgical technique that requires a patient to be awake and co-operative for all or part of an intracranial procedure. Securing an airway with a device can be necessary during these procedures electively (ie, for an asleep-awake-asleep anesthetic approach) or emergently.[1,2] In a recent systematic review of 4971 patients undergoing AC, the rate of unanticipated emergency conversion to general anesthesia that required an invasive airway was 2%.[2] Anesthesiologists must recognize that the emergent need for an airway device can occur at any stage of AC, and be prepared to secure the airway without delay.

Supraglottic airway devices are commonly used as the primary intervention for emergency airway management[3] because of their accessibility and ease of insertion. Laryngeal mask airway (LMA) placement can fail in up to 2% of cases, regardless of the type of surgery.[4,5] For patients undergoing AC, the rate of LMA failure might be significantly higher, as emergent placement can be impeded by the challenges of a partially flexed and fixed head in stereotactic pins, an exposed brain, and impeded access to the airway by surgical drapes (Fig. 1).[6]

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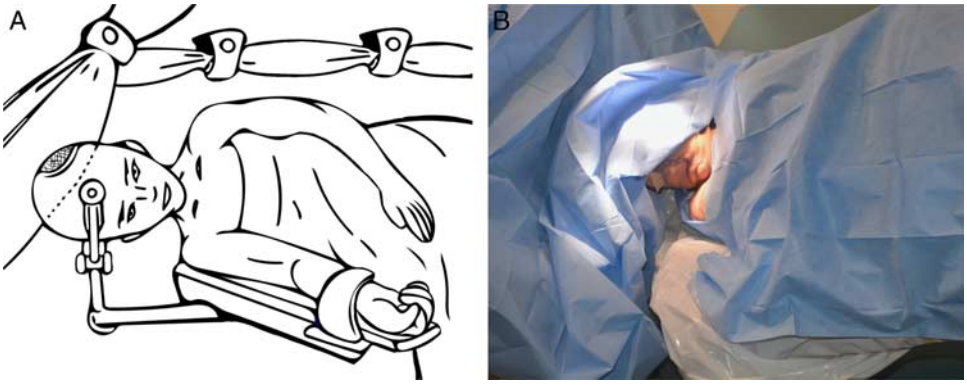


FIGURE 1. Patient positioning for awake craniotomy. A, Schematic representation of a patient's positioning during an awake craniotomy. Emergency airway management in patients undergoing awake craniotomy has several unique challenges: the patient's head is fixed in stereotactic pins with the head partially flexed, the brain is exposed, and the access to the patient's airway is impeded by surgical drapes. B, The cadaver was positioned during the airway simulation. full color online

To date, there are no dedicated studies that have assessed the best approach for emergently securing an airway device in patients undergoing AC. Therefore, the objective of this study was to compare the feasibility of airway management using an LMA and 4 alternative airway management techniques in a cadaver positioned to simulate the conditions of AC. We hypothesized that an LMA would be the most feasible approach in patients in this position, followed by video-assisted orotracheal intubation.

METHODS

To simulate typical AC airway management conditions, we placed a defrosted human cadaver in the right lateral position with a 3-pin skull fixation (Fig. 1). The cadaver in this study was purchased from a vendor who ensured that the study design was in agreement with the donor's wishes. The experiment was conducted in the simulation center at the Mayo Clinic in Jacksonville, FL.

All certified registered nurse anesthetists (CRNAs) and anesthesiologists at the Mayo Clinic in Jacksonville, FL, were invited to participate in the study. After providing informed consent, 9 participants were administered a questionnaire that assessed their current profession (CRNA or anesthesiologist), years in practice, whether their current clinical practice included AC, their first choice for securing the airway among the available devices, and their next approach if an LMA was in place and ventilation was not possible (whether or not an LMA was their first choice for airway management).

Each participant was instructed to secure the airway in the cadaver simulation with 5 airway management devices in random order. The airway management modalities were as follows: LMA (I-Gel Intersurgical, East Syracuse, NY), video laryngoscopy (C-MAC; Karl Storz Endoscopy, El Segundo, CA) using the regular Macintosh style blade, oral fiberoptic bronchoscopy (FOB), oral FOB through a pre-existing LMA, and nasal FOB (Karl Storz Endoscopy). Participants had access to an assistant to aid in airway management when requested. The time to successful LMA placement (defined as the ability to deliver positive pressure manual ventilation without a leak at 10 cm of H_2O) and time to vocal cord visualization were recorded. After the simulation, participants rated which device was easiest to use and ranked the devices in order of preference for emergent airway management in a patient undergoing AC. Each participant was assigned a unique number for blinded analysis.

Statistical analysis was performed with SPSS (version 1.0.0.1347) for Mac OS. Non-normal distribution of data were presented as the median and interquartile range (IQR) and nonparametric tests were used (Mann-Whitney and Kruskal-Wallis tests) to compare the variables. Proportions were compared with a z score. Data were considered statistically significant between groups when P-values were < 0.05.

RESULTS

Preference of Airway Management Device Before Simulation

The participants' characteristics are described in Table 1. The questionnaire administered before the simulation demonstrated that 8 of the participants chose an LMA and 1 participant chose video-assisted (C-MAC) orotracheal intubation as their first choice for emergency airway management during AC. Five participants selected video-assisted (C-MAC) orotracheal intubation as their next option after a failed LMA placement; the remaining participants chose FOB-assisted intubation through a pre-existing LMA (2 participants), FOB-assisted orotracheal intubation (1 participant), or FOB-assisted nasotracheal intubation (1 participant).

Simulation Results

During the simulation, only 3 approaches resulted in successfully securing an airway device for 100% of participants (Fig. 2): LMA (median time to secure the airway 6 s; IQR, 5 to 10 s), FOB-assisted intubation through an LMA (median time to secure the airway 41s; IQR, 23 to 51 s), and video laryngoscopy-assisted (C-MAC) orotracheal intubation

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TABLE 1. Characteristics of the Participants and Time to LMA Placement or Vocal Cord Visualization With Each Device

Participants' Characteristics (n)	LMA	Oral FOB Through LMA	Video Laryngoscopy	Oral FOB	Nasal FOB
Profession					
Anesthesiologist (5)	*5 (1)*	*23 (12)*	43 (94)	170 (49)	96 (59.5)
CRNA (4)	*11.5 (12.5)*	*55 (85.5)*	138.5 (200.7)	NA	NA
No. years in practice (y)					
< 5 (2)	31 (18)	138.5 (73.5)	179.5 (130.5)	NA	NA
5-9 (2)	5.5 (0.5)	32 (19)	119.5 (7.5)	NA	NA
10-14 (2)	7 (1)	34 (11)	135.5 (92.5)	NA	NA
> 20-24 (3)	5 (3)	30 (9.5)	18 (19)	169 (49)	117 (57.5)
Current practice in awake craniotomy					
Yes (7)	8 (6)	45 (26.5)	112 (131)	139 (25)	50 (35)
No (2)	5 (1)	26.5 (3.5)	24.5 (18.5)	NA	118.5 (43.5)

Nine individuals with advanced airway skills participated in the simulation. The time to LMA placement or vocal cord visualization is shown with respect to the participants' profession (anesthesiologist or CRNA), number of years in practice, and whether their current practice includes awake craniotomy procedures.

Data are presented as median (IQR) seconds and statistically significant differences are italicized.

CRNA indicates certified registered nurse anesthetists; FOB, fiberoptic bronchoscopy; IQR, interquartile range; LMA, laryngeal mask airway; NA, not applicable (< 3 participants were successful with that device, and median and IQR could not be calculated).

(median time to secure the airway 49 s; IQR, 43 to 127 s). However, the oral and nasal FOB approaches demonstrated success rates of only 44.4% (median time to secure the airway 154.5 s; IQR, 134.25 to 182 s) and 55.6% (median time to secure the airway 75 s; IQR, 50 to 117 s), respectively.

The time to secure the airway using the 5 modalities was compared (Fig. 2). An LMA was the fastest method ($P = 0.001$), and FOB-assisted intubation through an LMA was faster than FOB-assisted orotracheal intubation ($P = 0.021$). There were no significant differences in time to secure the airway among the other airway management devices. No statistically significant differences were found between years in practice and time to successful LMA placement or successful vocal cord visualization. However, anesthesiologists secured the airway faster than CRNAs with an LMA (median. 5 and 11.5 s, respectively; $P = 0.014$) and with FOB-assisted intubation through an LMA (median, 23 and 55 s, respectively; $P = 0.05$).

Preference of Airway Management Device After Simulation

After the simulation, 100% of participants reported that an LMA would be their first choice for emergent airway management during AC, followed by FOB-assisted intubation through an LMA (7 participants) or a video laryngoscope (2 participants) in the event that an LMA failed to properly seat. Compared with the approaches preferred before the simulation, after the simulation more participants chose FOB intubation through an LMA as their preferred approach if LMA failed to properly seat (2 participants before the simulation vs. 7 after the simulation; $P = 0.018$) (Fig. 2).

DISCUSSION

This is the first study to simulate airway management conditions and compare airway management approaches for AC using a human cadaver in a 3-pin skull fixation. Our principle findings were that: (1) an LMA was the fastest and most reliable method to secure the airway; (2) whereas most participants preferred the use of an LMA before the simulation, all participants preferred that method after the simulation; (3) FOB-assisted intubation through an LMA was the second-fastest approach for airway management; (4) whereas few participants before the simulation preferred FOB-assisted intubation through an LMA if LMA failed, a significantly higher proportion of participants identified this method as their next preferred choice after the simulation; and (5) the success of several airway management approaches was highly variable among participants.

Conversion of AC to general anesthesia and emergency placement of an invasive airway device is infrequent, but can have devastating consequences if securing the airway proves challenging or impossible. During AC, unique airway challenges can make airway management particularly difficult. Thus, simulation of these conditions is an effective method to learn which approaches are more feasible than others. For example, Brockerville et al[7] used a mannequin with its head fixed in stereotactic headframe to simulate airway management conditions for patients undergoing placement of a deep brain stimulator. Similar to our results, participants took longer to secure the airway using a video laryngoscope compared with an LMA (Portex Soft Seal laryngeal mask).

The cadaver simulation changed the participants' perceptions of which airway management device might be most useful in the event of a failed LMA placement. Historically, the FOB was the preferred method to facilitate orotracheal intubation in patients with a suspected difficult airway.[8,9] As the advent of the video laryngoscope, many clinicians prefer this approach over FOB-assisted intubation for difficult airways because of its relative ease of use, portability, and improved laryngoscopic view compared with direct laryngoscopy.[10] Therefore, most anesthesiologists today are highly skilled in the use of video laryngoscopy, and would choose this method to facilitate emergency intubation in an anticipated difficult airway scenario.[11]

Before the simulation, the majority of participants selected a video-assisted orotracheal intubation for airway management if an LMA failed to properly seat. The use of video laryngoscope during the simulation proved challenging

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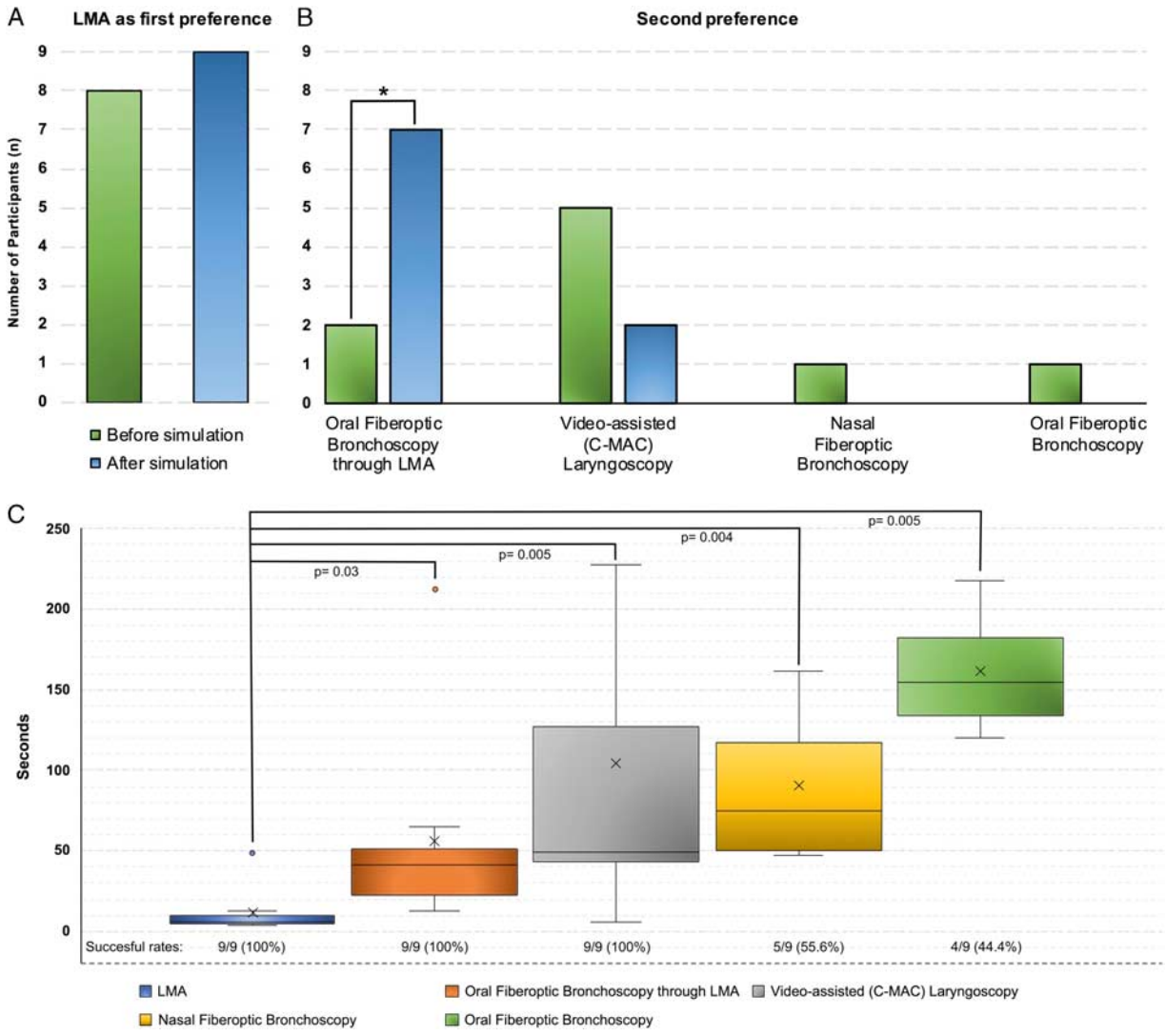


FIGURE 2. Participants' first (A) and second (B) preferences for securing a patient's airway with the head fixed in pins and the body positioned lateral before and after the cadaver simulation; and success rates and times to secure the cadaver's airway with each device (C). A, Before the simulation, 8 of 9 participants chose a laryngeal mask airway (LMA) to secure the airway emergently during an awake craniotomy. After the simulation, all participants chose an LMA as their first choice in emergency airway management. B, Compared to before the simulation, significantly more participants chose fiberoptic bronchoscopy through an LMA as their second option in the event that the LMA failed to properly seat after the simulation (*$P < 0.05$). C, Placement of an LMA was faster than all the other airway devices tested ($P < 0.05$), and demonstrated the lowest variability among participants. By contrast, video laryngoscopy and fiberoptic bronchoscope-assisted orotracheal and nasotracheal intubation demonstrated a high rate of variability among participants. Three devices had a 100% success rate: LMA, fiberoptic bronchoscopy (FOB) through the LMA, and video laryngoscopy. The rates of successful oral and nasal FOB were each <60%. full color online

for many participants, however, and this approach demonstrated a high rate of variability. When the head is fixed and partially flexed for patients undergoing AC, both video laryngoscope and FOB-assisted intubation can be especially challenging because of partial airway obstruction from the tissues in the posterior pharynx. Displacement of the posterior pharyngeal tissue is difficult with either approach individually, as our simulation demonstrated. Unsuccessful intubation attempts (especially in the presence of posterior airway obstruction) can result in significant oropharyngeal trauma and bleeding, reducing intubation success rate.[12]

Although the use of FOB and video laryngoscopy might be preferred to secure a difficult airway in both elective and emergency settings, this approach might not be best for patients during AC. Alternatively, the presence of an LMA, even 1 that is not sealed appropriately to deliver positive pressure ventilation, can significantly displace the tissue in the posterior pharynx and improve the likelihood of successful FOB-assisted intubation when performed through the LMA.[13] During our simulation, most participants demonstrated a high success rate using FOB-assisted intubation through an LMA, which became

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the preferred approach for most participants after an LMA failed to seat.

In an emergency situation when an endotracheal tube is necessary and a FOB is not immediately available, blind insertion of an endotracheal tube through a correctly positioned LMA results in successful intubation on the first attempt $>50\%$ of the time.[14] The use of a FOB to intubate through the LMA would greatly increase the likelihood of successful intubation. We recommend that a FOB be immediately available during AC and that anesthesia providers familiarize themselves with this equipment to prepare for emergency implementation.

This study had several limitations. The small sample size reflected our concern that the integrity of the airway tissues in the cadaver would break down with repeated airway manipulation and therefore hinder future intubation attempts. Moreover, although we studied only 5 airway approaches, several additional commercially available approaches (including different types of video laryngoscopes and LMAs) should be examined more fully. Furthermore, it is impossible to precisely simulate the conditions of a real patient undergoing AC. Although cadavers present a more realistic simulation than mannequins, defrosted airway tissues represent an imperfect model of live human tissue. For ethical and practical reasons, it would not have been feasible to perform this study in live patients. We recognize that using a cadaver makes the simulation of adequate oxygenation and ventilation difficult, and, as in a real patient, a seemingly well-placed LMA might not deliver effective positive pressure ventilation. In these patients, a definitive airway through an endotracheal tube might still be necessary.[15] Moreover, vocal cord visualization was chosen as a comparator of LMA placement in order to preserve the cadaver's tissue, although the time to vocal cord visualization does not guarantee successful intubation. Lastly, although we demonstrated that anesthesiologists were quicker than CRNAs in securing the airway with FOB-assisted bronchoscope through the LMA, the generalizability of this isolated finding is limited because of inherent differences in training, experience, and practice with the use of FOB among anesthesiologists and CRNAs within and between institutions.

In conclusion, these results support our hypothesis that an LMA is the fastest and most reliable primary method to secure an airway in a laterally positioned cadaver with 3-pin skull fixation. Recognizing that an LMA insertion might fail and securing an airway through an endotracheal tube can become necessary, we recommend that FOB and video laryngoscope airway equipment should be readily available during AC, with a clear backup plan including the size and length of the endotracheal tube through the LMA and whether or not to remove the LMA after successful intubation. An attempt to visualize the vocal cords through the LMA using FOB should be attempted before removing the LMA for alternative techniques.

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REFERENCES

1. Gruenbaum SE, Meng L, Bilotta F. Recent trends in the anesthetic management of craniotomy for supratentorial tumor resection. *Curr Opin Anaesthesiol*. 2016;29:552–557. doi: 10.1097/ACO.0000000000000365
2. Stevanovic A, Rossaint R, Veldeman M, et al. Anaesthesia management for awake craniotomy: systematic review and meta-analysis. *PLoS One*. 2016;11:e0156448. doi: 10.1371/journal.pone.0156448
3. Meng L, McDonagh DL, Berger MS, et al. Anesthesia for awake craniotomy: a how-to guide for the occasional practitioner. *Can J Anaesth*. 2017;64:517–529. doi: 10.1007/s12630-017-0840-1
4. Ramachandran SK, Mathis MR, Tremper KK, et al. Predictors and clinical outcomes from failed Laryngeal Mask Airway Unique: a study of 15,795 patients. *Anesthesiology*. 2012;116:1217–1226. doi: 10.1097/ALN.0b013e318255e6ab
5. Nagaro T, Yorozuya T, Sotani M, et al. Survey of patients whose lungs could not be ventilated and whose trachea could not be intubated in university hospitals in Japan. *J Anesth*. 2003;17:232–240. doi: 10.1007/s00540-003-0187-3
6. Deras P, Moulinie G, Maldonado IL, et al. Intermittent general anesthesia with controlled ventilation for asleep-awake-asleep brain surgery: a prospective series of 140 gliomas in eloquent areas. *Neurosurgery*. 2012;71:764–771. doi: 10.1227/NEU.0b013e3182647ab8
7. Brockerville M, Unger Z, Rowland NC, et al. Airway management with a stereotactic headframe in situ-a mannequin study. *J Neurosurg Anesthesiol*. 2018;30:44–48. doi: 10.1097/ANA.0000000000000402
8. Zlotnik A, Gruenbaum SE, Gruenbaum BF, et al. Awake fiberoptic intubation and general anesthesia in a parturient with mirror syndrome and a predicted difficult airway. *Isr Med Assoc J*. 2011; 13:640–642. PMID: 22097238
9. Gruenbaum SE, Gruenbaum BF, Tsaregorodtsev S, et al. Novel use of an exchange catheter to facilitate intubation with an Aintree catheter in a tall patient with a predicted difficult airway: a case report. *J Med Case Rep*. 2012;6:108. doi: 10.1186/1752-1947-6-108
10. Amaniti A, Papakonstantinou P, Gkinas D, et al. Comparison of laryngoscopic views between C-MAC and conventional laryngoscopy in patients with multiple preoperative prognostic criteria of difficult intubation. An observational cross-sectional study. *Medicina (Kaunas)*. 2019;55:1–12. doi: 10.3390/medicina55120760
11. Holmes MG, Dagal A, Feinstein BA, et al. Airway management practice in adults with an unstable cervical spine: the harborview medical center experience. *Anesth Analg*. 2018;127:450–454. doi: 10.1213/ANE.0000000000003374
12. Grabert J, Klaschik S, Guresir A, et al. Supraglottic devices for airway management in awake craniotomy. *Medicine (Baltimore)*. 2019;98:e17473. doi: 10.1097/MD.0000000000017473
13. Endigeri A, Ganeshnavar A, Varaprasad B, et al. Comparison of success rate of BlockBuster (R) versus Fastrach (R) LMA as conduit for blind endotracheal intubation: a prospective randomised trial. *Indian J Anaesth*. 2019;63:988–994. doi: 10.4103/ija.IJA_396_19
14. Campbell RL, Biddle C, Assudmi N, et al. Fiberoptic assessment of laryngeal mask airway placement: blind insertion versus direct visual epiglottoscopy. *J Oral Maxillofac Surg*. 2004;62:1108–1113. doi: 10.1016/j.joms.2003.10.014
15. Joffe AM, Aziz MF, Posner KL, et al. Management of difficult tracheal intubation: a closed claims analysis. *Anesthesiology*. 2019;131: 818–829. doi: 10.1097/ALN.0000000000002815

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